

System Design Document

Project: Cebuana Lhuillier Parking Reservation System
Version: 2.0
Date: February 24, 2026
Related: [SRS](#) | [Architecture Diagrams](#) | [ERD](#) | [Sequences](#) | [Compliance](#)

1. Technology Stack

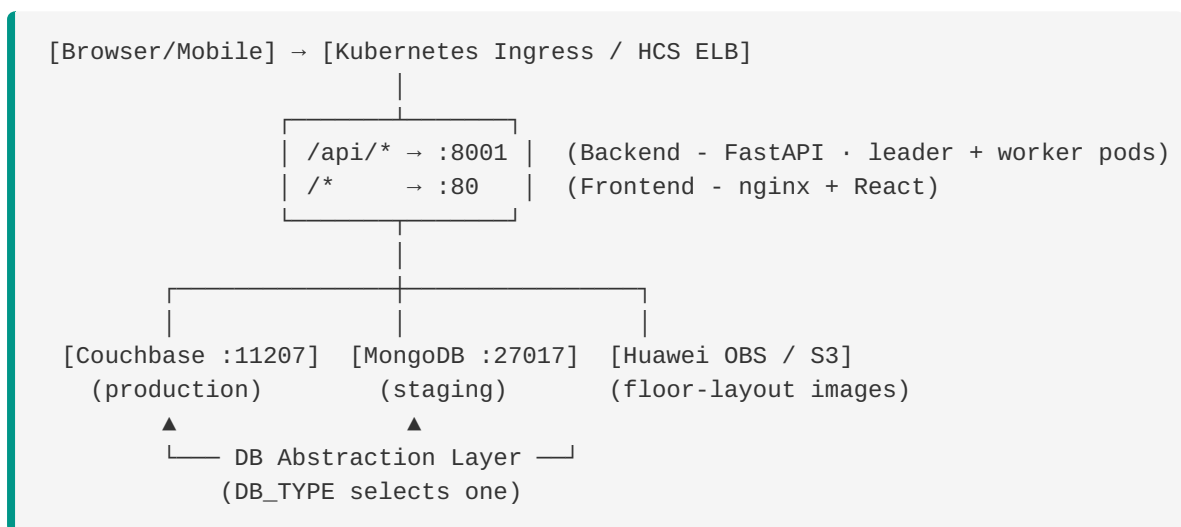
Layer	Technology	Version	Purpose
Frontend	React	19.0	SPA framework
UI Components	shadcn/ui (Radix)	Latest	Accessible component library
Styling	TailwindCSS	3.x	Utility-first CSS
Charts	Recharts	3.6	Dashboard visualizations
Routing	React Router	7.5	Client-side navigation
HTTP Client	Axios	1.8	API communication
Backend	FastAPI	0.110	Async Python API framework
ORM/Driver	Motor (Mongo) · couchbase-py	3.x · 4.x	Async DB drivers, selected via DB_TYPE
Auth	PyJWT + bcrypt	Latest	Token generation + password hashing
Database	MongoDB or Couchbase	7.x / Server 7 (Capella or Enterprise)	Pluggable via DB Abstraction Layer
Storage	Local FS or Huawei OBS (boto3 / S3)	1.42 (boto3)	Pluggable via Storage Abstraction Layer
AI	Emergent LLM SDK (optional, behind AI_INSIGHTS_ENABLED)	0.1	AI insights generation
QR	qrcode + Pillow	Latest	QR code generation

Layer	Technology	Version	Purpose
Feature flags	features.py (env-driven, public / api/system/features)	—	AI Insights · Attendant Mode · Self Check-In Window

2. High-Level Architecture

See [Architecture Diagrams](#) for visual representation.

2.1 Component Architecture



2.2 Module Structure

Backend (Python/FastAPI):

```

backend/
├── server.py          # App orchestrator (middleware, startup, routes, RUN_BACKGROUND_TASKS)
├── config.py          # Environment config + logging
├── features.py        # Runtime feature flags (AI Insights, Attendant Mode, Self Check-In Wi
├── database/          # DB Abstraction Layer (DI selected by DB_TYPE)
│   ├── interface.py   # Async Mongo-like API surface
│   ├── mongodb.py     # Motor-backed adapter
│   └── couchbase_db.py # Couchbase SDK adapter (Capella + Enterprise)
├── storage/           # Storage Abstraction Layer (DI selected by STORAGE_TYPE)
│   ├── interface.py   # put/get/delete/exists
│   ├── local.py       # Local volume (UPLOAD_DIR)
│   └── obs.py         # Huawei OBS / S3 (boto3 path-style + s3v4)
├── auth/
│   └── security.py     # JWT, bcrypt, role guards, session fingerprint
├── models/            # Pydantic request/response models (incl. building_policy, parking_con
├── routes/            # API endpoint handlers (15 modules – see system.py for /system/featur
│   ├── reservations.py # Booking, cancel, confirm, QR, self check-in, sticker checks
│   ├── system.py       # /db-info, /features (public), /collection-stats, /sync
│   ├── waitlist.py     # Waitlist + notify_next_waitlisted_user
│   └── ... (others)
├── services/
│   ├── background.py   # Auto no-show + self-checkin window sweep + immediate slot release + v
│   └── notifications.py
└── scripts/
    ├── db_cleanup.py
    ├── migrate_mongo_to_couchbase.py
    └── build_defect_validation_response.py

```

Frontend (React):

```

frontend/src/
├── App.js                # Routes, ProtectedRoute, layout (includes /admin/event-blocking)
├── context/
│   └── AuthContext.js   # Auth state, login/logout, session
├── services/
│   ├── api.js           # Axios instance, all API methods
│   └── eventBlocksAPI.js # Event blocks API methods (NEW)
├── components/
│   ├── ui/              # shadcn/ui components
│   ├── NotificationBell.js
│   └── ForgotPassword.jsx # Forgot password modal (NEW)
├── pages/
│   ├── LoginPage.js     # Login with forgot password link + uncontrolled password inputs
│   ├── UserDashboard.js
│   ├── BookingPage.js
│   ├── VehiclesPage.js
│   ├── ReservationsPage.js
│   ├── ProfilePage.js   # Includes change password section
│   ├── admin/
│   │   ├── AdminLayout.js
│   │   ├── AdminDashboard.jsx
│   │   ├── UserManagement.js
│   │   ├── BuildingManagement.jsx
│   │   ├── ZoneManagement.js
│   │   ├── ParkingConfig.js # Includes main_building_exclusive toggle
│   │   ├── AdminReservations.jsx # Includes cancellation reason dialog
│   │   ├── EventBlocking.jsx # Event blocking management (NEW)
│   │   ├── AttendantManagement.js
│   │   ├── Reports.js
│   │   └── Settings.js
│   ├── auth/
│   │   └── ForcePasswordChange.jsx # Uncontrolled password inputs (CWE-312 fix)
│   └── attendant/
│       └── AttendantDashboard.js
└── lib/
    └── utils.js         # cn() helper

```

3. Data Architecture

See [ERD](#) for entity-relationship diagram.

3.1 Collections Summary

Collection	Records	Key Fields	Indexes
users	38	id, email, password, role, main_building, tags, company, job_family	id (unique), email (unique)

Collection	Records	Key Fields	Indexes
sessions	108	id, user_id, fingerprint, revoked	id (unique), user_id
reservations	45	id, user_id, slot_id, building_id, date, status, qr_token, no_show_at, slot_released	id, user_id, building_id, (slot_id+date+status), qr_token
buildings	6	id, name, address	id (unique)
floors	12	id, label, building_id	id (unique), building_id
parking_slots	121	id, label, floor_id, building_id, status	id (unique), floor_id
vehicles	10	id, user_id, plate_number, make, model, color	id (unique), user_id, plate_number (unique)
zones	3	id, name, building_ids, user_ids	id (unique)
notifications	10	id, user_id, title, message, type, read	id (unique), (user_id+created_at)
parking_configs	3	id, building_id, release_time, booking_window_days, main_building_exclusive	building_id (unique)
ai_insights	1	id, content, generated_by, generated_at	id, generated_at
site_content	1	key, heading_line1, description, badges	key (unique)
building_policies	—	id, building_id, policy_type, enabled, max_users_per_slot, open_floor_ids, requires_sticker	id (unique), building_id (unique)
slot_registrations	—	id, slot_id, building_id, floor_id, user_id, vehicle_plate, sticker_number, status	id (unique), (slot_id+user_id+status)
waitlist_entries	—	id, user_id, building_id, preferred_date, preferred_start_time,	id (unique), (building_id+preferred_date+status)

Collection	Records	Key Fields	Indexes
		preferred_end_time, status, position, notified_at	
event_blocks	—	id, event_name, reason, building_id, floor_id, slot_ids[], date, start_time, end_time, reservation_ids[], created_by, created_at	id (unique), building_id
migrations	1	key, ran_at	—

3.2 Data Flow

User Action → React Component → Axios (withCredentials)
 → K8s Ingress (/api prefix) → FastAPI Router
 → Auth Middleware (JWT + Session + Fingerprint)
 → Route Handler → DB Abstraction Layer (Motor OR Couchbase SDK) → DB
 → Pydantic Response Model → JSON → React State → UI Update

4. Integration Architecture

Integration	Type	Purpose	Authentication
MongoDB	Database (staging)	Document store via Motor	<code>MONGO_URL</code> connection string
Couchbase Capella / Enterprise	Database (production)	Document store via couchbase-py	<code>COUCHBASE_CONNECTION_STRING</code> + user/pass + Trust Store (TLS)
Huawei OBS / S3	Object storage	Floor-layout images	AK/SK pair (<code>OBS_ACCESS_KEY</code> / <code>OBS_SECRET_KEY</code>)
Emergent LLM API	External AI (optional)	AI parking insights	API key (<code>EMERGENT_LLM_KEY</code>), gated by <code>AI_INSIGHTS_ENABLED</code>
File System	Storage (staging)	Floor plan images (local volume)	Local disk (<code>UPLOAD_DIR</code>)

4.1 External API Integration

AI Insights (Emergent LLM, optional):

```

Admin triggers → POST /api/reports/ai-insights
  → features.ai_insights_enabled() guard
  → If disabled → 503 immediately (UI hides the surface entirely)
  → Else: build prompt with parking stats
  → emergentintegrations.llm.chat.LlmChat
  → Stream response → Save to ai_insights collection
  → Return markdown to frontend

```

4.2 Feature-Flag Surface

The public endpoint `GET /api/system/features` returns the live runtime state:

```

{
  "ai_insights_enabled": true,
  "attendant_mode_enabled": true,
  "self_checkin_window_minutes": 15
}

```

Read by the React `useFeatures` hook once at SPA load and cached for the tab lifetime. Used to hide optional surfaces (AI Insights, Self Check-In button) so a deployment running with a feature off shows no traces of it in the UI.

5. Security Architecture

5.1 Authentication Flow

1. User submits credentials → `POST /api/auth/login`
2. Backend validates password (bcrypt) → creates JWT token (30min expiry)
3. JWT stored in HttpOnly secure cookie (not accessible via JS)
4. Session record created in DB with device fingerprint (User-Agent hash)
5. Every subsequent request: cookie sent automatically → JWT validated → session checked in DB → fingerprint verified
6. On failure: 401 response → frontend redirects to login

5.2 Authorization Model

Guard	Implementation	Routes
<code>get_current_user</code>	JWT + session + fingerprint validation	All authenticated routes
<code>require_admin</code>	<code>get_current_user + role == 'admin'</code> check	User CRUD, reports, config, site content

Guard	Implementation	Routes
<code>require_attendant</code>	<code>get_current_user + role in ['attendant', 'admin']</code>	Daily reservations, no-show, QR scan

5.3 Security Headers

Header	Value
Strict-Transport-Security	max-age=31536000; includeSubDomains
Content-Security-Policy	default-src 'self'; script-src 'self' 'unsafe-inline' 'unsafe-eval'; ...
Permissions-Policy	camera=(), microphone=(), geolocation=()
X-Content-Type-Options	nosniff
X-Frame-Options	DENY
X-XSS-Protection	1; mode=block
Referrer-Policy	strict-origin-when-cross-origin

6. Business Architecture

6.1 Process Map

EMPLOYEE ONBOARDING:

Admin bulk uploads CSV → Users created → Zone assigned → Main building set
→ Employee receives credentials → First login → Change password → Ready to book

DAILY BOOKING FLOW:

Employee opens app → Dashboard → "Book Now" → Select building/floor/date/slot
→ View hourly timeline to verify availability → Confirm → QR code generated → Notification :
→ (If fully booked: Join waitlist → Notified when slot opens → Book within window)

PARKING OPERATIONS:

Attendant Mode (ATTENDANT_MODE_ENABLED=true, default):

Attendant opens daily view → Sees reservations → Car arrives → Scans QR / Confirms
→ No-show after cutoff → Reports no-show → no_show_at timestamp recorded
→ User notified → Slot released X min after no_show_at (per config)
→ Waitlist triggered for the freed slot

Self Check-In Mode (ATTENDANT_MODE_ENABLED=false):

Parker arrives → opens "My Reservations" → green "Check in" button visible
→ Taps button (within start_time + SELF_CHECKIN_WINDOW_MINUTES, default 15)
→ Status flips to confirmed
→ If button NOT tapped within window → background loop flips to no_show
→ Slot released IMMEDIATELY → Waitlist triggered → no_show_count++ → in-app notice

EVENT BLOCKING:

Admin navigates to Event Blocking → Selects building/floor/date/time/slots
→ Names the event + provides reason → Submits → System creates confirmed reservations
→ Blocked slots appear as unavailable to regular users during the event period
→ Admin can delete event block to release slots when event is over

ADMIN CANCELLATION:

Admin views reservations → Clicks Cancel → Modal prompts for cancellation reason
→ Admin submits reason → Reservation cancelled → Parker notified with reason
→ Waitlist triggered for freed slot

PASSWORD MANAGEMENT:

User forgets password → Clicks "Forgot Password?" on login → Enters email → Admin notified
→ Admin resets password → User logs in with new password → Prompted to change again
OR
User wants to change password → Profile page → Enters current + new password → Saved

REPORTING:

Admin opens Reports → Views stats/charts → Filters by date/building
→ Generates AI insight → Saves for history → Shares with management

7. Diagrams Index

Diagram	Type	Location
Component Architecture	UML Component	architecture.md
Deployment Architecture (CCE)	UML Deployment	architecture.md
Application Layer Detail	UML Component	architecture.md
Frontend Component Tree	UML Component	architecture.md
Entity Relationship	ERD	erd.md
Login Sequence	UML Sequence	sequences.md
Booking Sequence	UML Sequence	sequences.md
Attendant No-Show Sequence	UML Sequence	sequences.md
Admin Bulk Upload Sequence	UML Sequence	sequences.md
QR Scan Verification Sequence	UML Sequence	sequences.md
Self Check-In Sequence	UML Sequence	sequences.md
Background No-Show + Waitlist Sequence	UML Sequence	sequences.md
Waitlist Notification & Claim Sequence	UML Sequence	sequences.md
Session Validation Sequence	UML Sequence	sequences.md
Compliance Matrix	Traceability	compliance-matrix.md

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